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Digital Pet

1. **Why does the pet screen need state, and which data could remain in immutable widget configuration(**Distinguish changing values from configuration and explain the widget/State relationship**)?**

The pet screen needs state because _happiness, _hunger, _energy, _petName, _gameOver, and _ hasWon change. The data that can remain in immutable widget configuration is the StatefulWidget itself.

2. **What does setState() do, and what does it not do(**Describe the callback and scheduled rebuild; it is not permanent storage and does not itself save data**)?**

setState() does the scheduled rebuild of the state object's widget tree. setState() does not permanently save the different data like _happiness, and _hunger.

3. **How does BlendMode.modulate combine source and filter colors, and what image works best(**Discuss per-channel multiplication and why dark source pixels can make a tint hard to see**)?**

BlendMode.modulate combines source and filter colors by multiplying the color components.

4. **Why must the periodic timer be canceled in dispose() (**Explain lifecycle ownership and the risk of callbacks after a screen is removed**)?**

The periodic timer must be canceled because it belongs to the State object, so while the screen is active, it controls the pet. After the screen is removed, Flutter calls dispose(), and the State object can't call setState().

5. **What are the trade-offs of tinting one image instead of storing separate mood images(**Consider package size, visual quality, asset flexibility, and when distinct artwork is still appropriate**)?**

A trade-off of using one image instead of storing separate mood images is that if the pet's expression doesn't change, the visual quality may not be appealing.

6. **Why can Flutter rebuild a widget after setState() without recreating the whole app(**Explain declarative UI and that Flutter updates the affected element/render tree where possible**)?**

Declarative UI is where Flutter does not have to tell all the individual elements what to change. Flutter compares new widgets with the existing widgets so setState() doesn't recreate everything, just wha† is needed.

7. **How would you separate pet rules from the widget tree as the app grows(**Give one concrete boundary between presentation, state, and game logic**)?**

Since I would separate between Presentation, state, and game logic, I would first have PetHomePage holding the widgets, buttons, image, and layout. Next, the State would hold happiness, hunger, energy, petName, hasWon, and gameOver. Lastly, game logic would hold all the rules of the game in a different class.

8. **How does Dart null safety help with a mood-color function or optional timer(**Explain non-null return types and why nullable resources should be checked or safely canceled**)?**

Non-null return values don't return null; they return a value instead, like a color in my app. In the code, I used the ? on the nullable values and made it so they are cancellable.

9. **When would ColorFiltered be the wrong choice for a pet's mood or evolution(**Discuss when each mood needs distinct artwork, shapes, or more than a color tint**)?**

The color filter modifies the color of the image I used. It does not change the expressions in the image. For my code I would meed a different picture for each mode pet_happy, pet_sad, pet_sleepy